NimbusEdge Technologies Pvt. Ltd. Network Routing Incident Analysis

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Introduction

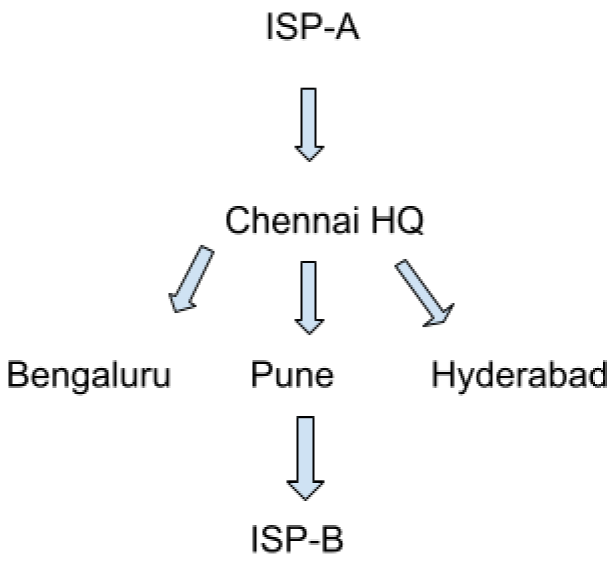
Content:

NimbusEdge Technologies Pvt. Ltd. experienced a major network outage on March 18, 2024. The outage affected multiple branch offices including Chennai, Bengaluru, Pune, and Hyderabad. Several enterprise clients faced service disruptions due to routing issues involving RIP v2, IGRP, OSPF, and BGP protocols. This report analyzes the incident, identifies the root causes, examines the impact on business operations, and provides recommendations to prevent similar incidents in the future.

Network Topology Overview

Content:

The NimbusEdge network consists of Chennai Headquarters, Bengaluru, Pune, Hyderabad, and the Coimbatore Disaster Recovery Site. Different routing protocols such as RIP v2, IGRP, OSPF, and BGP were used across the network. ISP-A served as the primary internet connection, while ISP-B acted as the backup connection. The mixed routing environment increased network complexity and contributed to the outage.



Incident Timeline

TIME	EVENT
09:42	Client ticket raised
10:15	Delivery subnet missing
11:05	High latency reported
12:30	OSPF issue identified
14:50	ISP-A fiber cut
15:40	Manual BGP failover started
16:20	ISP-B activated
18:55	Services restored

5. Root Cause Analysis

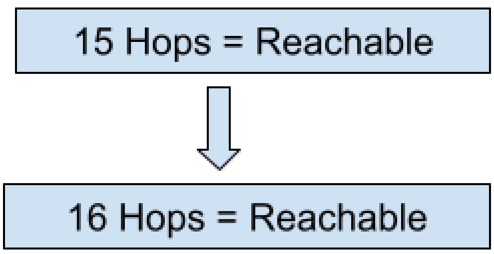
5.1 RIP v2 Issue

Explanation

The Delivery subnet was added to the network, but RIP v2 supports a maximum of 15 hops. Since the subnet was 16 hops away, RIP considered it unreachable and removed it from the routing table. As a result, users could not access the Delivery application.

Impact

- Delivery subnet became inaccessible.
- Application services stopped working.
- Users reported connectivity issues.



5.2 IGRP Issue

Explanation

The Bengaluru-Pune link became degraded. Although an alternate route was available, IGRP continued to use the slower path because of its metric calculation. This caused network congestion and delay.

Impact

- High network latency.
- Slow application performance.
- Increased user complaints.

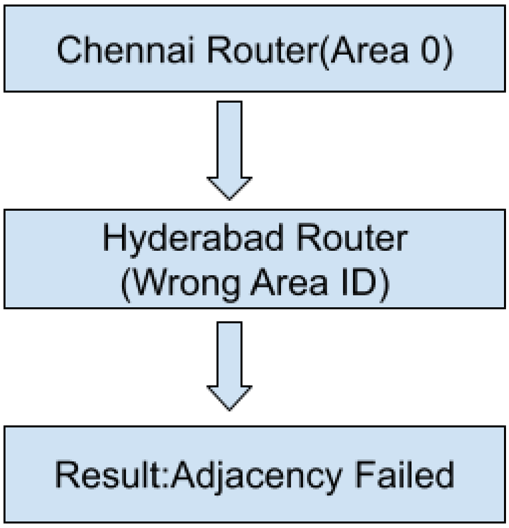
5.3 OSPF Issue

Explanation

A wrong Area ID was configured on the Hyderabad router. Because of this mismatch, OSPF adjacency could not be established with neighboring routers. Route information was not exchanged correctly.

Impact

- OSPF routes disappeared.
- Traffic could not reach destinations.
- Routing black hole occurred.



5.4 BGP Issue

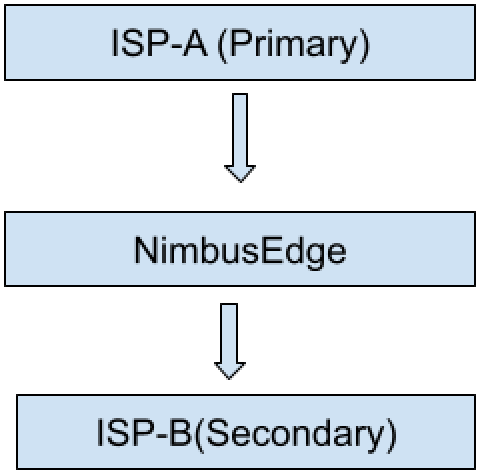
Explanation

The primary ISP-A connection failed due to a fiber cut. The backup ISP-B connection was available, but BGP failover policies were not configured correctly. Therefore, traffic did not automatically switch to ISP-B.

Impact

- Internet connectivity was interrupted.

- Network outage duration increased.
- Business services were affected.



Normal:ISP-A Active
Failure:ISP-B Active

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6. Business Impact

Explanation

The routing failure affected multiple enterprise customers and branch offices. Several business-critical applications became unavailable during the outage.

Effects

- Six enterprise clients affected.
- Service downtime lasted between 3 and 9 hours.
- SLA commitments were violated.
- Customer satisfaction decreased.
- Possible financial penalties occurred.

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7. Corrective Actions

Actions Taken

1. Static routes were added temporarily for affected networks.
2. IGRP metrics were adjusted to use a better route.
3. OSPF Area ID configuration was corrected.
4. BGP traffic was manually redirected to ISP-B.
5. Network services were monitored until full recovery.

8. Recommendations

Suggested Improvements

1. Replace RIP v2 with OSPF to avoid hop count limitations.
2. Migrate from IGRP to modern routing protocols such as EIGRP or OSPF.
3. Verify OSPF Area IDs before implementing configuration changes.
4. Regularly test BGP failover between ISP-A and ISP-B.
5. Implement automated network monitoring tools.
6. Perform routing audits periodically.
7. Follow strict change-management procedures.
8. Conduct disaster recovery and failover testing regularly.

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9. Conclusion

Conclusion

The NimbusEdge network outage was caused by multiple routing protocol issues involving RIP v2, IGRP, OSPF, and BGP. The RIP hop-count limitation, IGRP route selection problem, OSPF Area ID mismatch, and BGP failover failure collectively resulted in a major service disruption. The incident affected several enterprise clients and led to SLA violations. By adopting modern routing protocols, improving configuration management, and performing regular failover testing, similar incidents can be prevented in the future.

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10. References

1. Incident Report (NOC-INC-2024-031)
2. Network Topology Diagram
3. Root Cause Analysis (RCA) Report
4. NOC Change Log
5. Client SLA Breach Notice